



System architecture — *one source in, every format out*

A Gen-AI content transformation platform: the pipeline runs in the browser, and each model call goes to a cloud provider or to an on-device model, by the user's choice.

LEGEND

- **Private mode** · never leaves the device
- **Cloud mode** · AI requests
- **Accounts & data** · Supabase

USER'S DEVICE

BROWSER React 19 · TypeScript 5.7 · Vite 8 · Tailwind CSS 4 · installable PWA · the whole pipeline runs here

WORKSPACE UI

UI Source & configuration

Upload PDF · DOCX · image, paste text or a link. Pick from 10 formats or describe one; tone and audience.

Processing mode

Cloud APIs

Private

Hardware check (RAM · cores · GPU) and Ollama setup guide before Private runs.

UI Output canvas

Collects every draft; on-device drafts stream in token by token.

Compare up to 3 models side by side · refine one draft in plain words · Stop at any moment.

Export PowerPoint (.pptx), Markdown or text.

UI History · Analytics · Admin

Every draft with its format, model and word counts; usage totals.

Admin: all users and activity; disable, enable or delete accounts.

Every /workspace route sits behind an auth guard.

Document readers

PDF → text (pdf.js), DOCX → text (mammoth), TXT · CSV · JSON · MD. Scans → page images.

Image reader

Vision model (Gemini, Mistral) or on-device OCR (Tesseract.js). Always OCR in Private mode.

CLIENT-SIDE INGESTION

Link reader

Page → text through Jina Reader. Cloud mode only.

N1 Ingest node

Merges text, file, link, image

N2 Parse node

Local clean-up, no AI call

N3 Fit source

Size budget per provider

N4 Format × Model

One node per (format, model), fanned out in parallel: 2–3 per provider, 1 on-device.
tone · audience · instructions

N5 Translate

Whole text, chunk by chunk

N6 Refine router

Draft only, or draft + source

N7 Run controller

Smart generation check skips drafts you already have.
One AbortController per run: Stop keeps finished drafts.

N6 Route every call

Retries rate limits, 2-min timeout, own-key consent prompt before shared keys.

No own key → /api/chat, shared keys

Own API key → provider directly

Private mode → Ollama on this computer

MODEL GATEWAY · CHAT()



PRIVATE MODE Local engine · Ollama · qwen2.5vl:7b localhost:11434

Streamed /api/generate: drafts arrive token by token; Stop closes the stream at once.

num_ctx 4096 · temperature 0.3: fits an 8 GB laptop GPU, stays close to the source.

Zero data leaves: no server, no provider, no key. History keeps counts only.

CLOUD SERVICES

DB Supabase · accounts & data

Auth

Email sign-in and sessions. Disabled accounts are banned here and signed out everywhere.

Postgres + RLS

profiles · activity_logs. Each user reads only their own rows; the admin reads all.

Server-checked functions: admin_list_users · admin_set_user_disabled · admin_delete_user · my_account_disabled

WEB Jina Reader

Fetches a linked page server-side and returns it as text, because models can't open links themselves.

API Vercel

Static hosting

The React app, PWA manifest and service worker.

/api/chat function

Holds the shared provider keys in environment variables; rotates them and retries rate limits.

AI Cloud AI providers

11 drafting models

Groq

gpt-oss-120b (default) · gpt-oss-20b · qwen3.8-27b. Fastest drafts; the refine router.

Gemini

3.5 Flash-Lite · 3.8 Flash. Reads text and images; the longest sources.

Mistral

Large · Medium · Small · Nemo · Codestral. Vision on Medium and Small.

Users may paste their own key; it stays in the tab's memory and is sent only to that provider.

PRIVATE

In Private mode none of this column sees the document: no text, images, drafts or file names reach the cloud.

10 formats

+ any custom format, drafted in parallel

3 + 1 engines

Groq · Gemini · Mistral, or Ollama on-device

4,096 tokens

local window, sized for an 8 GB GPU

6 languages

full-length Indian-language translation

RLS on every table

data access enforced by the database

How it works

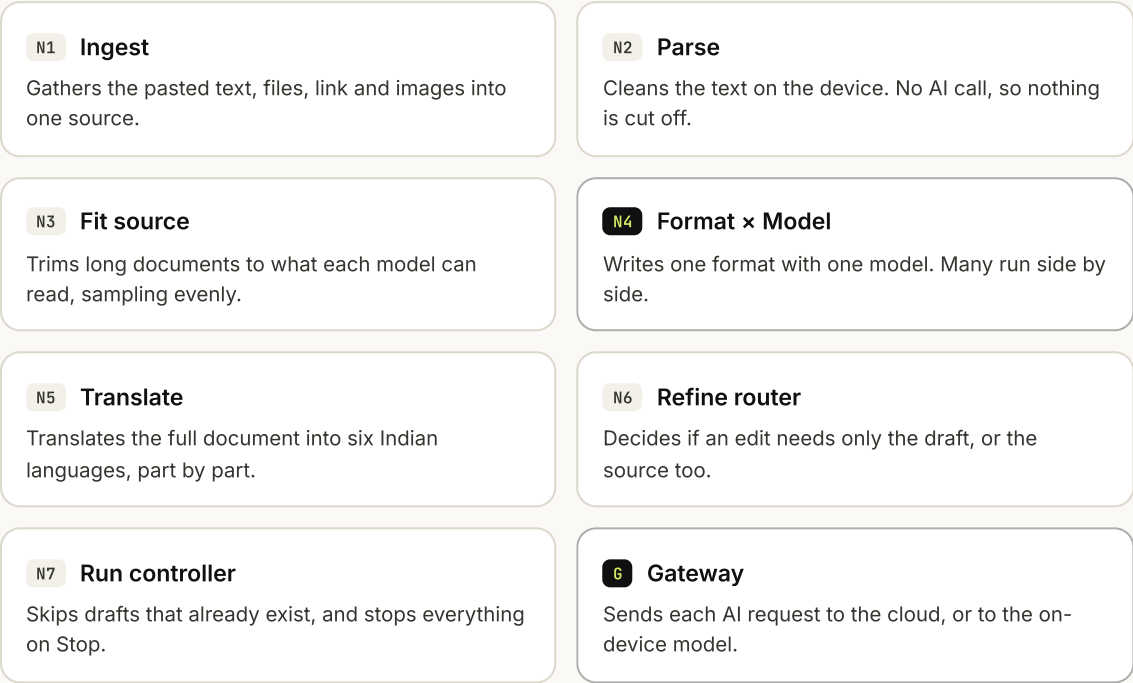
The whole pipeline runs in the user's browser. One gateway then sends each AI request either to a cloud provider, for speed, or to a model on the same computer, when a document must never leave it.

1 The journey of one request



2 What each node does

labels match page 1



3 Two ways to run

	Cloud APIs	Private, on-device
Who writes	Groq, Gemini, Mistral	qwen2.5vl:7b via Ollama
Your document	Sent to the provider	Never leaves the device
History keeps	An excerpt and drafts	Counts only
Best for	Speed and long sources	Confidential documents

4 Built-in safeguards

- ✓ **API keys stay on the server.** Shared keys never reach the browser.
- ✓ **Nothing switches silently.** If a user's key fails, we ask first.
- ✓ **Stop means stop.** One click cancels every request at once.
- ✓ **Each user sees only their own data.** The database enforces it.